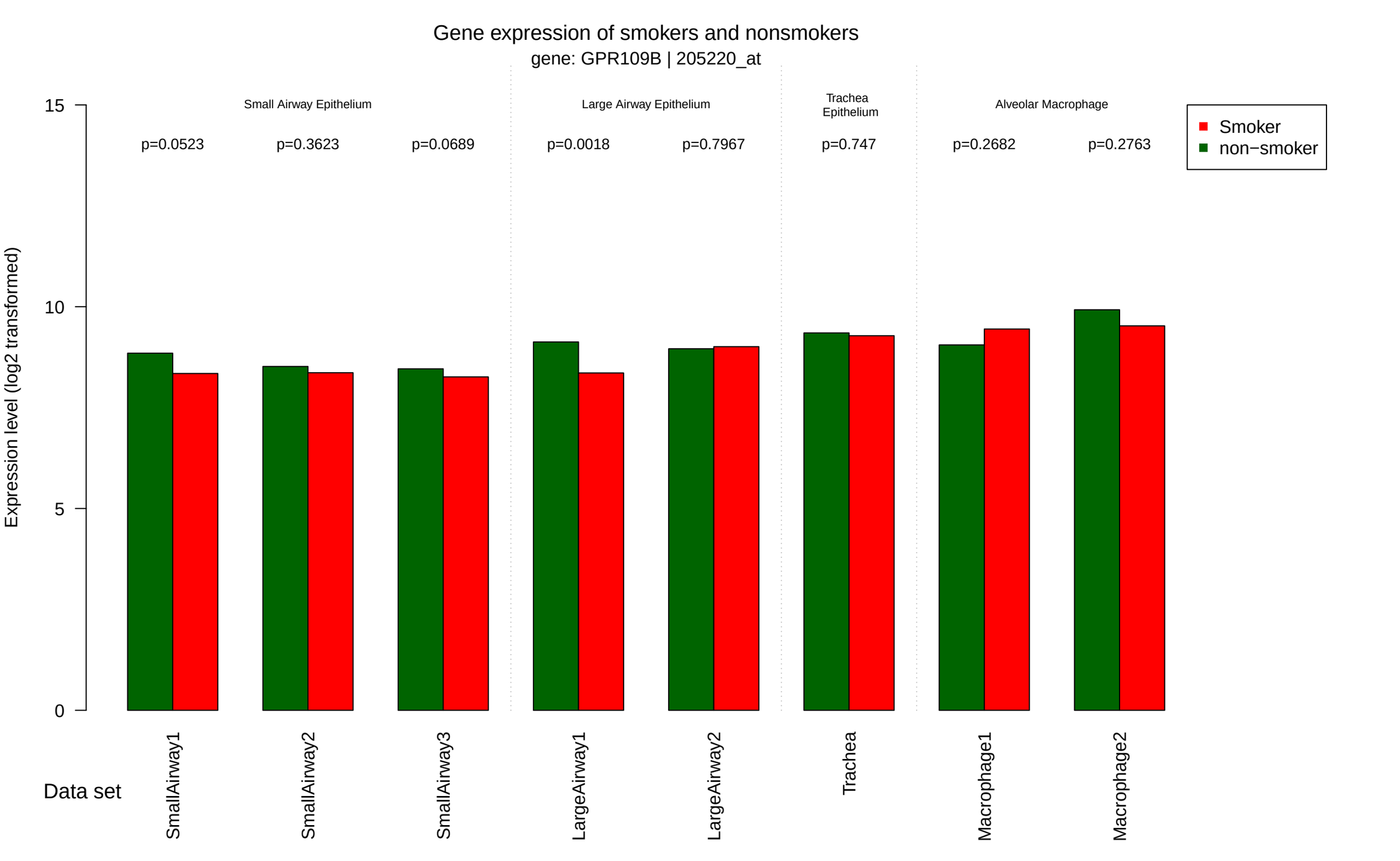
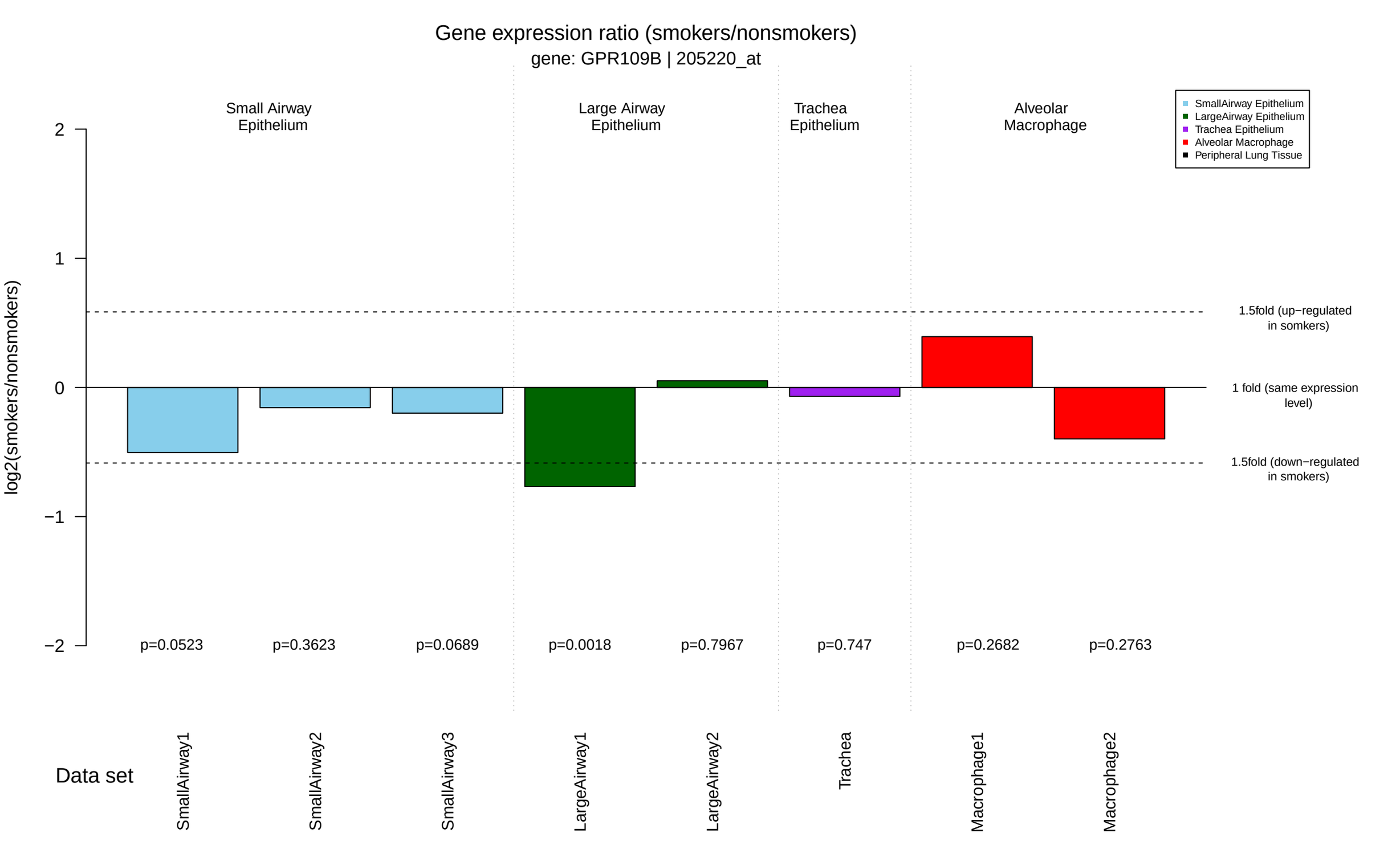
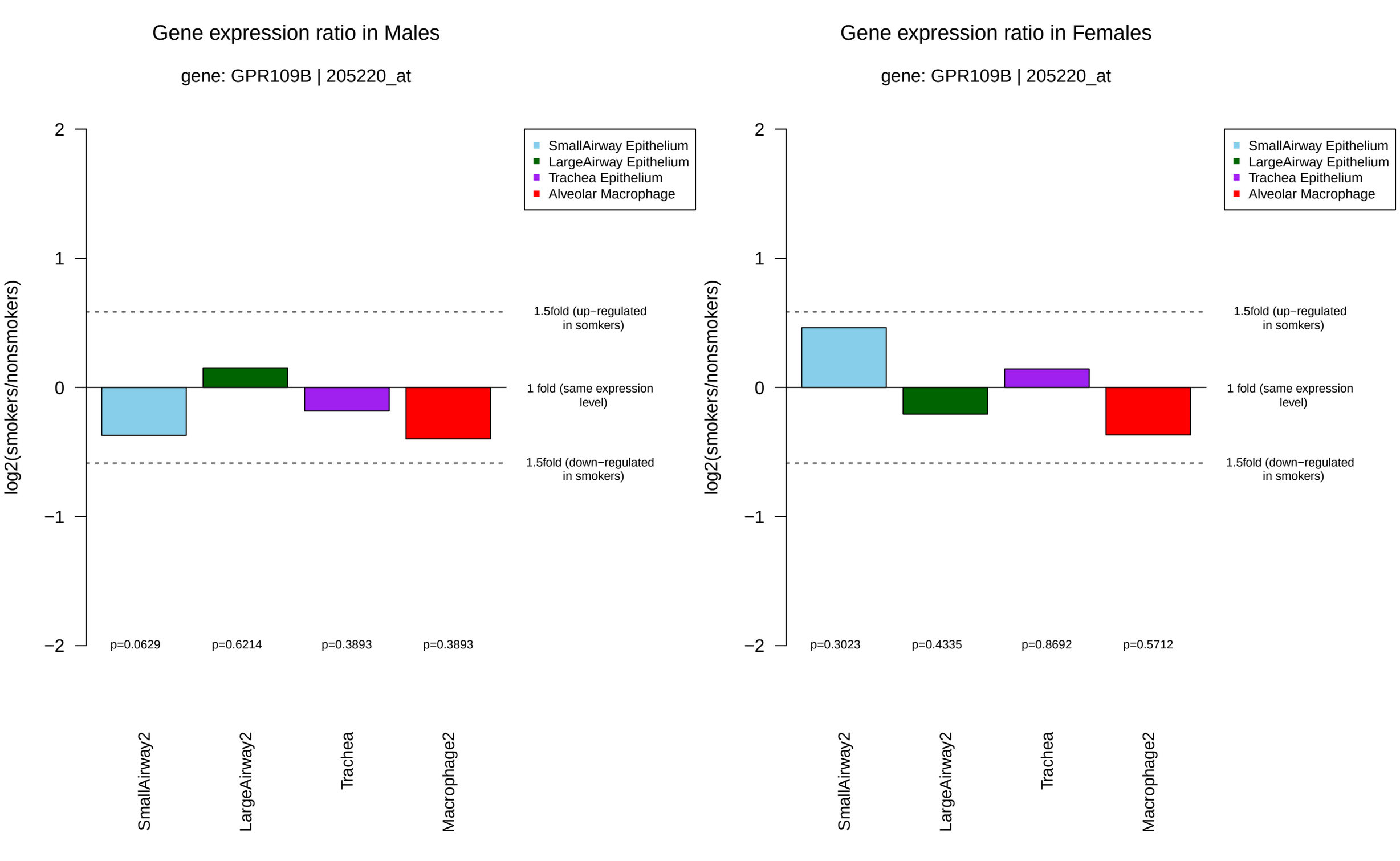
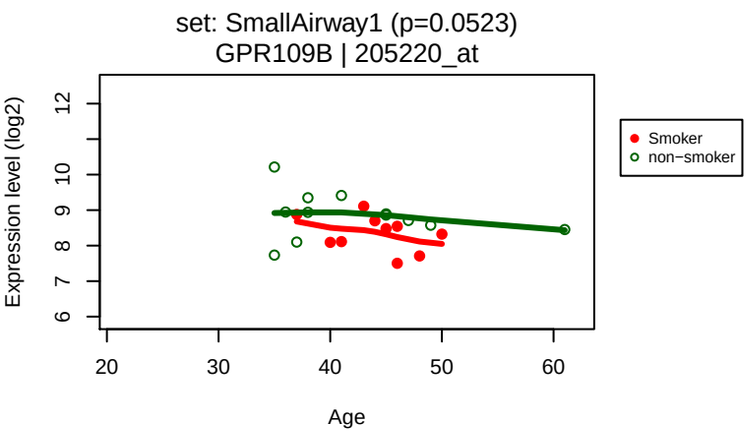
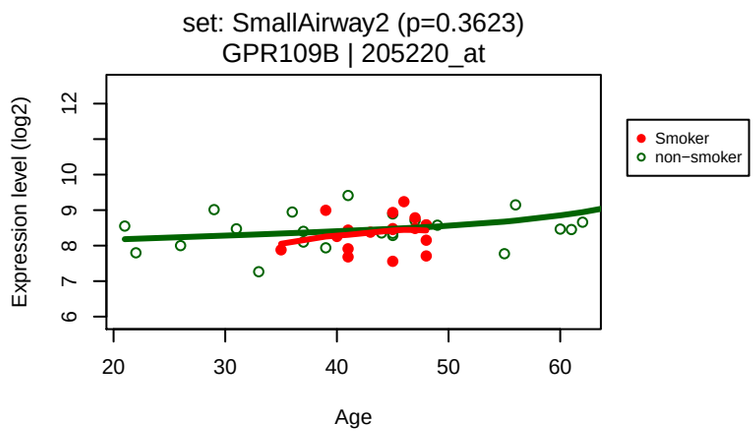
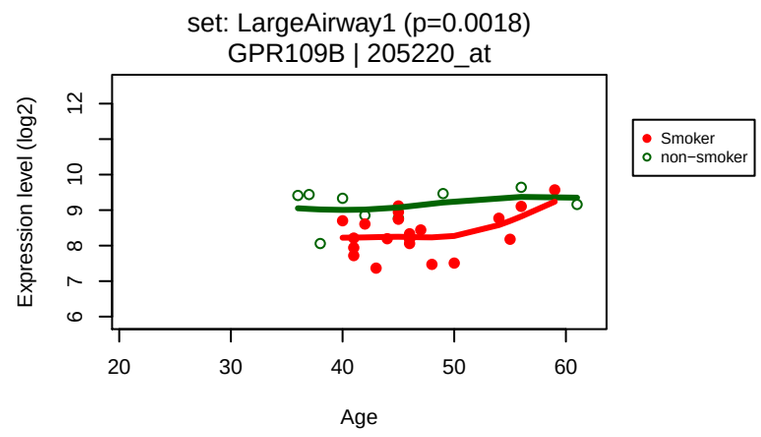

Gene expression of smokers and nonsmokers
gene: GPR109B | 205220_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0523
p=0.3623
p=0.0689
p=0.0018
p=0.7967
p=0.747
p=0.2682
p=0.2763
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

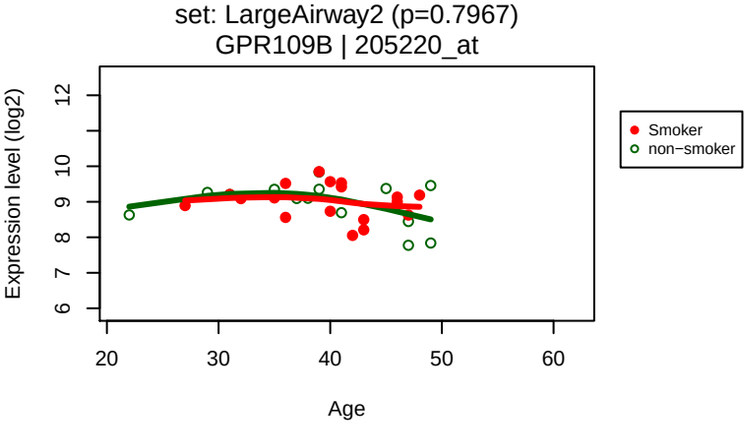

Gene expression ratio (smokers/nonsmokers)
gene: GPR109B | 205220_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0523
p=0.3623
p=0.0689
p=0.0018
p=0.7967
p=0.747
p=0.2682
p=0.2763
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

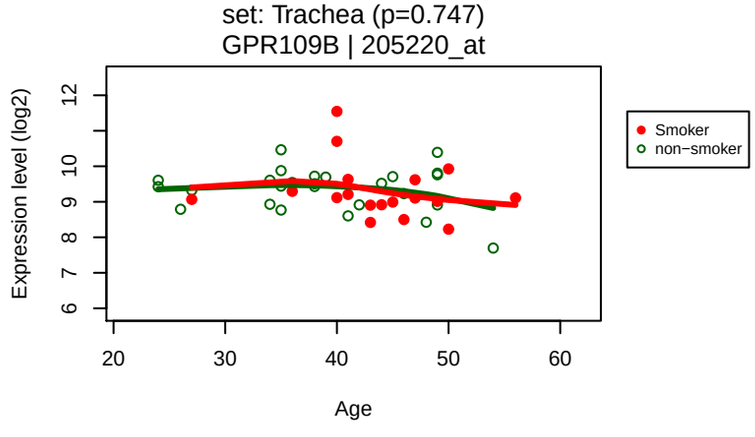

Gene expression ratio in Males
gene: GPR109B | 205220_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0629
p=0.6214
p=0.3893
p=0.3893
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPR109B | 205220_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3023
p=0.4335
p=0.8692
p=0.5712
SmallAirway2
LargeAirway2
Trachea
Macrophage2

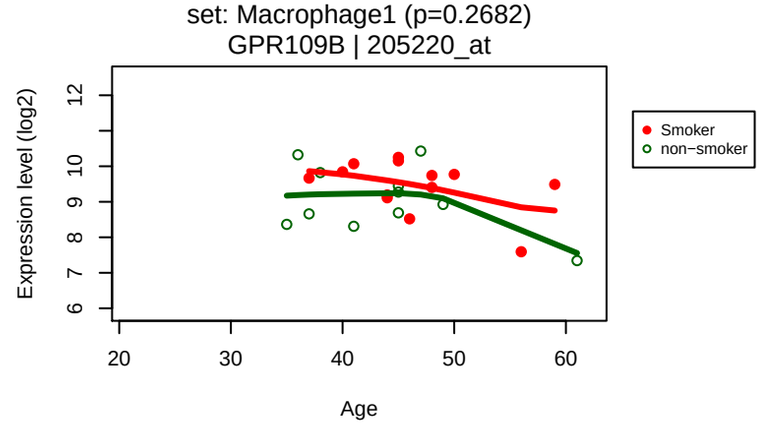

set: SmallAirway1 (p=0.0523)
GPR109B | 205220_at
Expression level (log2)
Age
Smoker
non−smoker

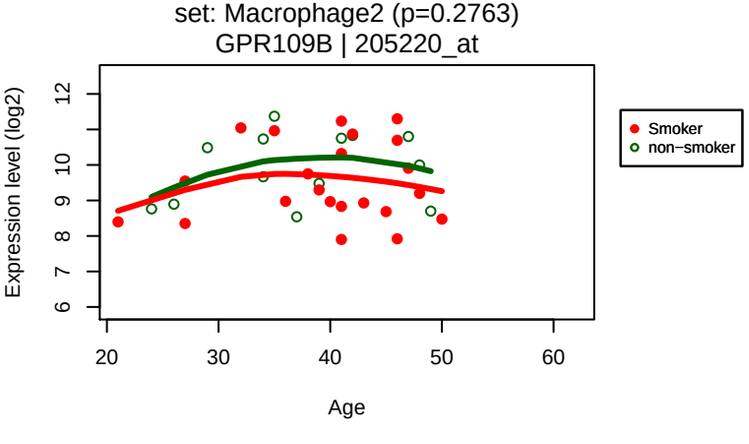

set: SmallAirway2 (p=0.3623)
GPR109B | 205220_at
Expression level (log2)
Age
Smoker
non−smoker

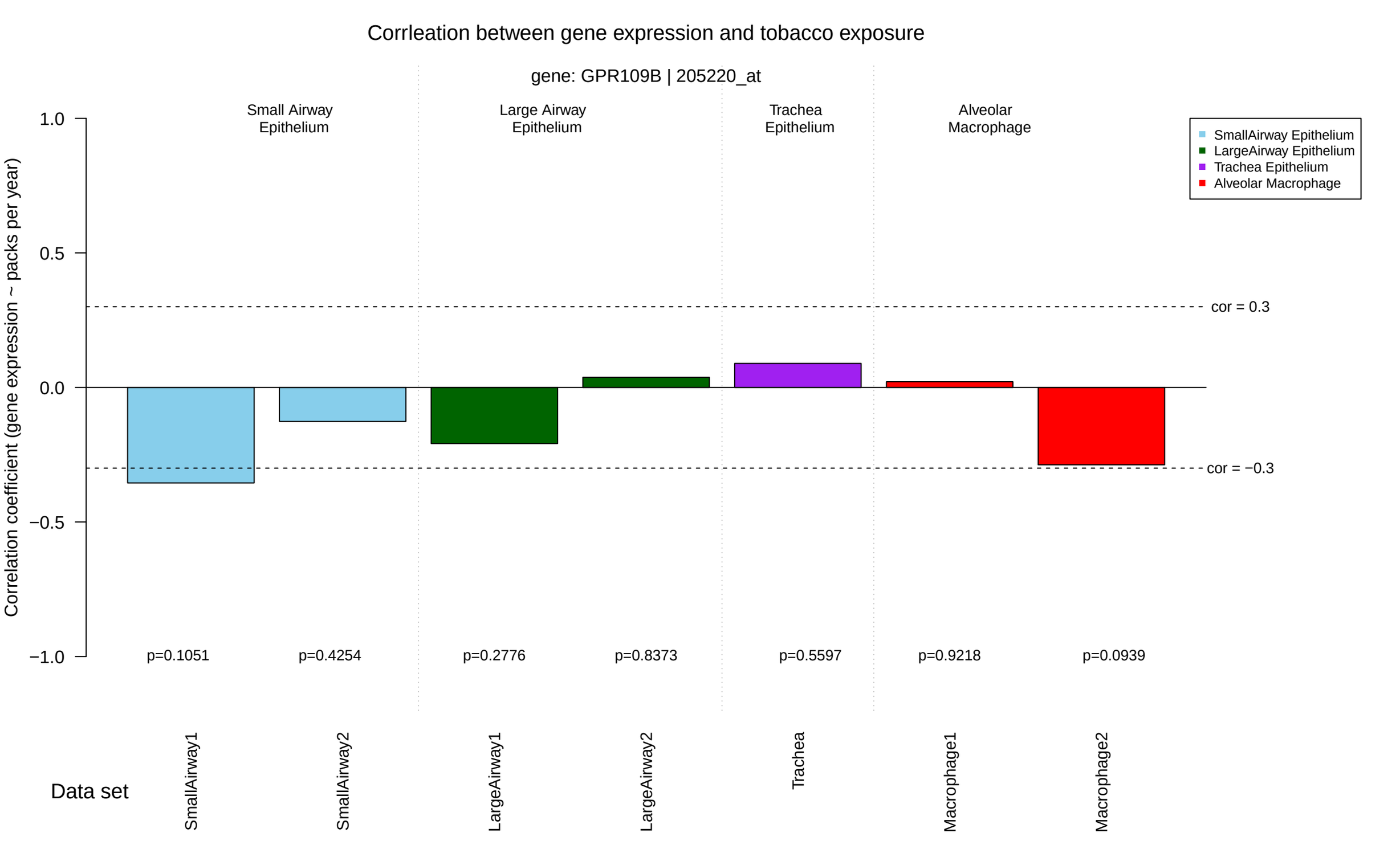

set: LargeAirway1 (p=0.0018)
GPR109B | 205220_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7967)
GPR109B | 205220_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.747)
GPR109B | 205220_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2682)
GPR109B | 205220_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2763)
GPR109B | 205220_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPR109B | 205220_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1051
p=0.4254
p=0.2776
p=0.8373
p=0.5597
p=0.9218
p=0.0939
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2